A. Course Handout (Version 1.0)

Institute/School Name	Chitkara University Institute of Engineering and Technology		
Department Name	Department of Computer Science & Engineering		
Programme Name	Bachelor of Engineering (B.E.), Computer Science & Engineering		
Course Name	Advanced System Design	Session	2025-2026
Course Code	23CS020	Semester/Batch	6th/2023
L-T-P (Per Week)	2-0-0	Course Credits	02
Pre-requisites	System Design(22CS024)	NHEQF Level	5.5
Course Coordinator	Dr. Suhasini	SDG Number	4,8,9

CLO01	Gain an understanding regarding principles, core concepts and architectures of large-scale and distributed systems.
CLO02	Apply coordination, consistency, and consensus algorithms in distributed systems
CLO03	Design secure and fault-tolerant data management architectures that scale using transactions, distributed storage, caching, and security controls
CLO04	Evaluate and design real time distributed applications using message queues, event streams, containerization, API management, techniques

1. **Objectives of the Course**

This course provides an in-depth understanding of the principles and practices involved in designing large-scale, distributed, and scalable systems for real-world applications. Students will develop the ability to analyse requirements, assess design trade-offs, and develop robust system architectures addressing performance, scalability, reliability, and security. The objectives of the course are:

i. To Gain an understanding regarding principles, core concepts and architectures of large-scale and distributed systems.
ii. To Apply coordination, consistency, and consensus algorithms in distributed systems
iii. To apply coordination, consistency, and consensus algorithms in distributed systems.
iv. To design robust, scalable, and secure distributed system architectures for practical use cases

2. **Course Learning Outcomes:** After completion of the course, student should be able to:

	Course Learning Outcome	*POs	**CL	***KC	Sessions
CLO01	Gain an understanding regarding principles, core concepts and architectures of large-scale and distributed systems.	PO1, PO2, PO5, PO12	K4	Factual, Conceptual, Procedural	14
CLO02	Apply coordination, consistency, and consensus algorithms in distributed systems	PO1, PO2, PO3, PO4, PO5, PO12	K4	Factual, Conceptual, Procedural	16
CLO03	Design secure and fault-tolerant data management architectures that scale using transactions, distributed storage, caching, and security controls.	PO1, PO2, PO3, PO4, PO5, PO12	K5	Factual, Conceptual, Procedural	20
CLO04	Evaluate and design real time distributed applications using message queues, event streams, containerization, API management, techniques.	PO1, PO2, PO3, PO4, PO5, PO6, PO9,PO1, PO11,PO12	K6	Factual, Conceptual, Procedural	10

Revised Bloom's Taxonomy Terminology
* PO's available at (shorturl.at/cryzF)
**Cognitive Level =CL
***Knowledge Categories = KC

Course Learning Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CLO01	H	M	–	–	L	–	–	–	–	–	–	L
CLO02	M	H	M	L	L	–	–	–	–	–	–	L
CLO03	M	H	H	M	M	–	–	–	–	–	–	L
CLO04	M	M	H	M	M	L	–	–	L	L	L	M

H=High, M=Medium, L=Low

3. ERISE Grid Mapping

Feature Enablement	Level (1-5, 5 being highest)
Entrepreneurship	3
Research	4
Innovation	4
Skills	5
Employability	5

4. Recommended Books:

B01: Coulouris, Dollimore, Kindberg, and Blair, Distributed Systems: Concepts and Design, 5th Edition, Pearson Education.

B02: Martin Kleppmann, Designing Data-Intensive Applications, O'Reilly Media, 2017.

B03: Andrew S. Tanenbaum and Maarten Van Steen, Distributed Systems: Principles and Paradigms, 2nd Edition, Pearson Education.

B04: Operating System Concepts Essentials, 10th Edition by Avi Silberschatz, Peter Galvin, Greg Gagne, Wiley Asia Student Edition, 2018.

B05: Database System Concepts", 6th Edition by Abraham Silberschatz, Henry F. Korth, S. Sudarshan, McGraw-Hill, 2010.

5. Other readings and relevant websites:

Serial No	Link of Journals, Magazines, websites and Research Papers
a)	https://system-design-handbook.pages.dev
b)	https://www.geeksforgeeks.org/system-design
c)	https://getvm.io/tutorials/distributed-systems-lecture-series
d)	https://github.com/rajeevranjancom/System_Design
e)	https://github.com/madd86/awesome-system-design
f)	https://www.youtube.com/watch?v=CuQmQpvw04I&t=32s

6. Recommended Tools and Platforms

Canva, Draw.io, Diagrams.net

7. Course Plan:

Lecture Number	Topics	Recommended Books/Resources
1-6	**Foundation of Large Scale Systems Design(Recap):** System design principles, Functional versus non-functional requirements, Latency vs throughput, Design Approaches and Architectural Patterns	Refer to Material for System Design (5th Sem.)
7-10	**Fundamentals of Distributed Systems:** Definition and characteristics of distributed systems, Advantages and challenges, Types of distributed systems	B01,B02,B03
11-14	Client–server and peer-to-peer models, Distributed System Failures: Failure models & fault tolerance	B01,B02,B03
15-20	**Advanced Distributed Systems Concepts:** Process coordination and Synchronization , Mutual exclusion, Deadlock Handling	B01,B04
21-24	Parallel Computing, Distributed Computing, Parallel Computing versus Distributed Computing	B01,B02,B03,B04
25-30	CAP theorem, PACELC theorem, Revision of Consistency Models, Consensus algorithms (Raft, Paxos – internal working), Leader election and failure detection	B01,B02,B03
	ST1(Syllabus Covered Lecture number 1-30)	
31-34	**Data Management and Security in System Design:** Basics of Transactions Processing, Conflict Serializability and Recoverability	B01,B05
35-40	Concurrency Control Techniques, Data Recovery Mechanisms, Data Security	B01,B05
41-44	**Advanced Distributed Data Management Concepts:** Distributed databases(Spanner, DynamoDB, Cassandra concepts), Data replication techniques, Data partitioning and Sharding	B01, B02,B03,B05
45-50	Distributed Transactions, Basics of File Systems, Distributed file systems (HDFS, GFS concepts), Distributed Cache, Security in Distributed Systems	B01, B02,B03,B05
51-56	**Advanced Distributed Architectural Concepts:** Distributed Message queues, Distributed Event Streams, Comparison of Message Queues and Event Streams, Containerization, Pub-Sub Design Patterns	B01,B02,B03
57-60	API Design and Versioning, Rate limiting & Throttling, API Gateways, Monitoring, Logging, Tracing in Distributed Scalable Systems, Capacity Planning and Incident Management	B01,B02,B03
ST-2 Syllabus (Lecture number 31-60)		
Full Semester(As Per topics covered in class)	**Real Time Application Design and Discussion(Using Tools like Draw.io etc.):** Sending Emails, Recommendation Systems, Data Post Processing, Real-time Chat Application, Distributed File Storage (Google Drive / Dropbox)	-
ETE (Syllabus = Lecture number 1-60)		

8. Delivery/Instructional Resources:

Lecture Numbers	Topics	Web References	Audio-Video References
1-6	**Foundation of Large Scale Systems Design(Recap):** System design principles, Functional versus non-functional requirements, Latency vs throughput, Design Approaches and Architectural Patterns	https://systemdesign.one/ https://www.geeksforgeeks.org/system-design	https://www.youtube.com/watch?v=CuQmQpvw04I&t=32s

7-10	**Fundamentals of Distributed Systems:** Definition and characteristics of distributed systems, Advantages and challenges, Types of distributed systems	https://www.geeksforgeeks.org/computer-networks/what-is-a-distributed-system/	https://www.youtube.com/watch?v=Azyizl9w2xo&list=PLrjkTql3jnm9FEOXHA_qjR-TMODlaIk-W
11-14	Client–server and peer-to-peer models, Distributed System Failures: Failure models & fault tolerance	https://www.geeksforgeeks.org/operating-systems/distributed-computing-system-models/	https://www.youtube.com/watch?v=Azyizl9w2xo&list=PLrjkTql3jnm9FEOXHA_qjR-TMODlaIk-W
15-20	**Advanced Distributed Systems Concepts:** Process coordination and Synchronization , Mutual exclusion, Deadlock Handling	https://www.geeksforgeeks.org/operating-systems/	https://www.youtube.com/watch?v=3Eaw1SSIqRg https://youtu.be/izS9J07ToRM?si=HAxcSYbDpDGT12Ig
21-24	Parallel Computing, Distributed Computing, Parallel Computing versus Distributed Computing	https://www.geeksforgeeks.org/dbms/difference-between-parallel-computing-and-distributed-computing/	https://youtu.be/FuUd5LrWWkE?si=bT9yCySMIWlIDbQO
25-30	CAP theorem, PACELC theorem, Revision of Consistency Models, Consensus algorithms (Raft, Paxos – internal working), Leader election and failure detection	https://systemdesign.one/ https://www.geeksforgeeks.org/dbms/the-cap-theorem-in-dbms/	https://www.youtube.com/watch?v=BaKtC-VIYrM
31-34	Data Management and Security in System Design: Basics of Transactions Processing, Conflict Serializability and Recoverability	https://www.geeksforgeeks.org/dbms/	https://youtube.com/playlist?list=PLxCzCOWd7aiFAN6I8CuViBuCdJgiOkT2Y&si=fdzMLdSh51RWgl8G
35-40	Concurrency Control Techniques, Data Recovery Mechanisms, Data Security	https://www.geeksforgeeks.org/dbms/	https://youtube.com/playlist?list=PLxCzCOWd7aiFAN6I8CuViBuCdJgiOkT2Y&si=fdzMLdSh51RWgl8G

41-44	Advanced Distributed Data Management Concepts: Distributed databases(Spanner, DynamoDB, Cassandra concepts), Data replication techniques, Data partitioning and Sharding	https://www.geeksforgeeks.org/dbms/distributed-database-system/	https://www.youtube.com/watch?v=BUOSup1APPw
45-50	Distributed Transactions, Basics of File Systems, Distributed file systems (HDFS, GFS concepts), Distributed Cache, Security in Distributed Systems	https://www.geeksforgeeks.org/distributed-systems/what-is-dfsdistributed-file-system/	https://www.youtube.com/watch?v=FtTbDjqZMRY
51-56	**Advanced Distributed Architectural Concepts:** Distributed Message queues, Distributed Event Streams, Comparison of Message Queues and Event Streams, Containerization, Pub-Sub Design Patterns	https://www.geeksforgeeks.org/system-design/message-queues-vs-event-streams-in-system-design/	https://www.youtube.com/watch?v=XvnppkWqJbs https://www.youtube.com/watch?v=tBXK49CB-BU
57-60	API Design and Versioning, Rate limiting & Throttling, API Gateways, Monitoring, Logging, Tracing in Distributed Scalable Systems, Capacity Planning and Incident Management	https://www.geeksforgeeks.org/system-design/api-throttling-vs-api-rate-limiting-system-design/	https://www.youtube.com/watch?v=MIJFyUPG4Z4

9. Action plan for different types of learners:

Slow Learners	Average Learners	Fast Learners
• Remedial Classes on Saturdays • Encouragement for improvement using Peer Tutoring • Use of Audio and Visual Materials • Use of Real-Life Examples	• Workshops • Formative Exercises used to highlight concepts and notions • E-notes and E-exercises to read ahead of the pedagogic material.	• Engaging students to hold hands of slow learners by creating a Peer Tutoring Group • Design solutions for complex problems • Design solutions for complex problems • Presentation on topics beyond those covered in CHO

10. Evaluation Scheme & Components: As per Choice Based Credit System(CBCS) , evaluation scheme given below:

Evaluation Component	Type of Component	No. of Assessments	Weightage of Component	Mode of Assessment
Component 1	Sessional Tests (STs)	02*	40%	Online(On-Campus)
Component 2	End Term Examination	01	60%	Online(On-Campus)
Total		**100%**		

* Out of 02 STs, Best 1 ST for final marks evaluation of STs will be considered.
**As per Academic Guidelines, a minimum of 75% attendance is required to become eligible for appearing in the End Semester Examination.

Syllabus of the Course:

Subject: Advanced System Design		
Topic (s)	**No. of Sessions**	**Weightage %**
Foundation of Large Scale Systems Design(Recap): System design principles, Functional versus non-functional requirements, Latency vs throughput, Design Approaches and Architectural Patterns, **Fundamentals of Distributed Systems:** Definition and characteristics of distributed systems, Advantages and challenges, Types of distributed systems, Client–server and peer-to-peer models, Distributed System Failures: Failure models & fault tolerance	**14**	**24%**
Advanced Distributed Systems Concepts: Process coordination and Synchronization , Mutual exclusion, Deadlock Handling, Parallel Computing, Distributed Computing, Parallel Computing versus Distributed Computing , CAP theorem, PACELC theorem, Revision of Consistency Models, Consensus algorithms (Raft, Paxos – internal working), Leader election and failure detection	**16**	**26%**
Data Management and Security in System Design: Basics of Transactions Processing, Conflict Serializability and Recoverability, Concurrency Control Techniques, Data Recovery Mechanisms, Data Security, **Advanced Distributed Data Management Concepts:** Distributed databases(Spanner, DynamoDB, Cassandra concepts), Data replication techniques, Data partitioning and Sharding, Distributed Transactions, Basics of File Systems, Distributed file systems (HDFS, GFS concepts), Distributed Cache, Security in Distributed Systems	**20**	**33%**
Advanced Distributed Architectural Concepts: Distributed Message queues, Distributed Event Streams, Comparison of Message Queues and Event Streams, Containerization, Pub-Sub Design Patterns, API Design and Versioning, Rate limiting & Throttling, API Gateways, Monitoring, Logging, Tracing in Distributed Scalable Systems, Capacity Planning and Incident Management	**10**	**17%**

This Document is approved by:

Designation	Name	Signature
Course Coordinator	Dr. Suhasini	
Head-Academic Delivery	Dr. Susheela Hooda	
Dean	Dr. Rupali Gill	
Date	30.01.2026	